

606023



Test Booklet No.

ENGLISH

Test Booklet Code

04

NEET RANKERS TEST SERIES-2025



Do not open this Test Booklet until you are asked to do so.

SYLLABUS:

Physics:

Ray Optics and Optical Instruments, Wave Optics, Dual Nature of Radiation and Matter, Atoms, Nuclei, Semiconductor Electronics: Materials, Devices and Simple Circuits

Chemistry:

Haloalkanes and Haloarenes, Alcohols, Phenols and Ethers, Aldehydes, Ketones and Carboxylic Acids, Amines, Biomolecules, Principles Related to Practical Chemistry

Botany:

Microbes in human welfare, Organisms and populations, Ecosystem, Biodiversity and conservation

Zoology:

Human Health and Diseases, Biotechnology Principles and Processes, Biotechnology and its Applications

Important Instructions:

- The Answer Sheet is inside this Test Booklet. When you are directed to open the Test Booklet, take out the Answer Sheet and fill in the particulars on ORIGINAL Copy carefully with blue/black ball point pen only.
- The test is of 3 hours 20 minutes duration and the Test Booklet contains 200 multiple-choice questions (four options with a single correct answer) from Physics, Chemistry and Biology (Botany and Zoology). 50 questions in each subject are divided into two Sections (A and B) as per details given below:
 - Section A shall consist of 35 (Thirty-five) Questions in each subject (Question Nos - 1 to 35, 51 to 85, 101 to 135 and 151 and 185). All questions are compulsory.
 - Section B shall consist of 15 (Fifteen) questions in each subject (Question Nos - 36 to 50, 86 to 100, 136 to 150, and 186 to 200). In Section B, a candidate needs to attempt any 10 (Ten) questions out of 15 (Fifteen) in each subject.

Candidates are advised to read all 15 questions in each subject of Section B before they start attempting the question paper. In the event of a candidate attempting more than ten questions, the first ten questions answered by the candidate shall be evaluated.
- Each question carries 4 marks. For each correct response, the candidate will get 4 marks. For each incorrect response, one mark will be deducted from the total scores. The maximum marks are 720.
- Use Blue/Black Ball Point Pen only for writing particulars on this page/mark responses on Answer Sheet.
- Rough work is to be done in the space provided for this purpose in the Test Booklet only.
- On completion of the test, the candidate must hand over the Answer Sheet (ORIGINAL and OFFICE Copy) to the Invigilator before leaving the Room/Hall. The candidates are allowed to take away this Test Booklet with them.
- The CODE for this Booklet is 04. Make sure that the CODE printed on the Original Copy of the Answer Sheet is the same as that on this Test Booklet. In case of discrepancy, the candidate should immediately report the matter to the Invigilator for replacement of both the Test Booklet and the Answer Sheet.
- The candidates should ensure that the Answer Sheet is not folded. Do not make any stray marks on the Answer Sheet. Do not write your Roll No. anywhere else except in the specified space in the Test Booklet/Answer Sheet.
- Use of white fluid for correction is NOT permissible on the Answer Sheet.
- Each candidate must show on-demand his/her Admit Card to the Invigilator
- No candidate, without special permission of the centre Superintendent or Invigilator, would leave his/her seat.
- The candidates should not leave the Examination Hall without handing over their Answer Sheet to the Invigilator on duty and sign (with time) the Attendance Sheet twice. Cases, where a candidate has not signed the Attendance Sheet second time, will be deemed not to have handed over the Answer Sheet and dealt with as an Unfair Means case.
- Use of Electronic/Manual Calculator is prohibited.
- The candidates are governed by all Rules and Regulations of the examination with regard to their conduct in the Examination Room/Hall. All cases of unfair means will be dealt with as per the Rules and Regulations of this examination.
- No part of the Test Booklet and Answer Sheet shall be detached under any circumstances.
- The candidates will write the Correct Test Booklet Code as given in the Test Booklet/Answer Sheet in the Attendance Sheet.
- Compensatory time of one hour five minutes will be provided for the examination of three hours and 20 minutes duration, whether such candidate (having a physical limitation to write) uses the facility of Scribe or not.

Name of the Candidate (in Capitals): _____

Roll Number : In figures _____

: In words _____

Centre of Examination (in Capitals): _____

Candidate's Signature: _____ Invigilator's Signature: _____

Facsimile Signature stamp of Centre Superintendent

Physics : Section-A (Q. No. 1 to 35)

1. An object is placed at a distance of 12 cm in front of a plane mirror. A virtual and erect image is formed by the mirror. Now, the mirror is moved by 4 cm towards the stationary object. The distance by which the position of image would be shifted, will be:

(1) 4 cm towards mirror
(2) 8 cm away from mirror
(3) 2 cm towards mirror
(4) 8 cm towards mirror

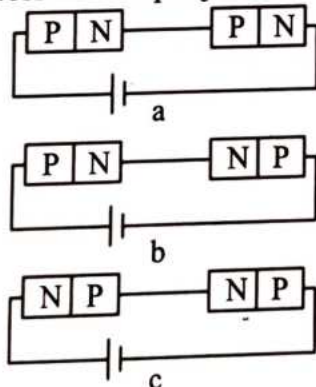
2. The ratio of the de-Broglie wavelength of proton and electron having same kinetic energy (Assume, $m_p = m_e \times 1849$)

(1) 1 : 43 (2) 1 : 62
(3) 1 : 30 (4) 2 : 43

3. A monochromatic light wave with wavelength λ_1 and frequency ν_1 in air enters another medium. If the angle of incidence and angle of refraction at the interface are 45° and 30° respectively, then the wavelength λ_2 and frequency ν_2 of the refracted wave are:

(1) $\lambda_2 = \lambda_1, \nu_2 = \sqrt{2}\nu_1$
(2) $\lambda_2 = \sqrt{2}\lambda_1, \nu_2 = \nu_1$
(3) $\lambda_2 = \frac{1}{\sqrt{2}}\lambda_1, \nu_2 = \nu_1$
(4) $\lambda_2 = \lambda_1, \nu_2 = \frac{1}{\sqrt{2}}\nu_1$

4. In the given circuits (a), (b) and (c), the potential drop across the two p-n junction are equal in:



- (1) Both circuit (a) and (c)
(2) Circuit (a) only
(3) Circuit (b) only
(4) Circuit (c) only

5. For a parallel beam of monochromatic light of wavelength ' λ ', diffraction is produced by a single slit whose width ' a ' is of the order of the wavelength of the light. If ' D ' is the distance of the screen from the slit, the width of the central maxima will be:

(1) $\frac{D\lambda}{a}$ (2) $\frac{Da}{\lambda}$
(3) $\frac{12D}{a}$ (4) $\frac{2D\lambda}{a}$

6. The difference between threshold wavelengths for two metal surface A and B having work function $\phi_A = 9 \text{ eV}$ and $\phi_B = 4.5 \text{ eV}$ in nm is:

(1) 264 (2) 276
(3) 540 (4) 138

7. The image formed by an objective of a compound microscope is:

(1) virtual and diminished
(2) real and diminished
(3) real and enlarged
(4) virtual and enlarged

8. The width of fringe is 2 mm on the screen in a double slit experiment for the light of wavelength of 400 nm. The width of the fringe for the light of wavelength 600 nm will be:

(1) 1.33 mm
(2) 2 mm
(3) 4 mm
(4) 3 mm

9. The momentum of an electron revolution in n th orbit is given by (Symbols have their usual meanings)

(1) $\frac{nh}{2\pi r}$ (2) $\frac{nh}{2r}$
(3) $\frac{nh}{2\pi}$ (4) $\frac{2\pi r}{nh}$

10. Two periodic waves of intensities I_1 and I_2 pass through a region at the same time in the same direction. The sum of the maximum and minimum intensities is:

(1) $2(I_1 + I_2)$ (2) $I_1 + I_2$
 (3) $(\sqrt{I_1} + I_2)^2$ (4) $(\sqrt{I_1} - I_2)^2$

11. Let K_1 and K_2 be the maximum kinetic energies of photo-electrons emitted when two monochromatic beams of wavelength λ_1 and λ_2 respectively are incident on a metallic surface. If $\lambda_1 = 3\lambda_2$, then

(1) $K_1 > \frac{K_2}{3}$ (2) $K_1 < \frac{K_2}{3}$
 (3) $K_1 = \frac{K_2}{3}$ (4) $K_2 = \frac{K_1}{3}$

12. A single slit of width a is illuminated by a monochromatic light of wavelength 600 nm. The value of a for which first minimum appears at $\theta = 30^\circ$ on the screen will be:

(1) $1.2 \mu\text{m}$ (2) $3 \mu\text{m}$
 (3) $0.6 \mu\text{m}$ (4) $1.8 \mu\text{m}$

13. The photodiode is used to detect the optical signals.

These diodes are preferably operated in reverse biased mode because;

- (1) fractional change in majority carries produce higher forward bias current.
 (2) fractional change in majority carriers produce higher reverse bias current.
 (3) fractional change in minority carriers produce higher forward bias current.
 (4) fractional change in minority carriers produce higher reverse bias current.

14. Barrier potential of a p-n junction diode does not depend on:

- (1) Diode design (2) Temperature
 (3) Forward bias (4) Doping density

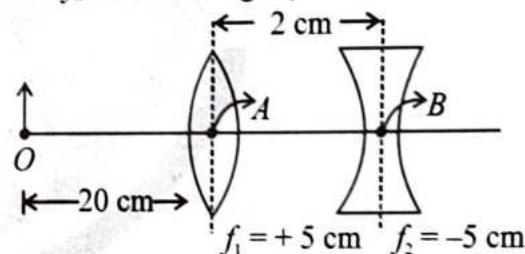
15. In photoelectric effect;

- A. The photocurrent is proportional to the intensity of the incident radiation.
 B. Maximum Kinetic energy with which photoelectrons are emitted depends on the intensity of incident light.
 C. Maximum K.E with which photoelectrons are emitted depends on the frequency of incident light.
 D. The emission of photoelectrons requires a minimum threshold intensity of incident radiation.
 E. Maximum K.E of the photoelectrons is independent of the frequency of the incident light.

Choose the **correct** answer from the options given below:

- (1) A and B only (2) B and C only
 (3) A and C only (4) A and E only

16. What is the position and nature of image formed by lens combination shown in figure? (where, f_1 and f_2 are focal lengths)

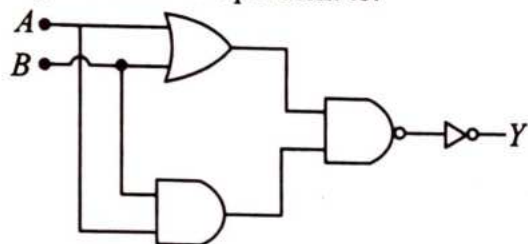


- (1) $\frac{20}{3}$ cm from point B at right, real
 (2) 70 cm from point B at right, real
 (3) 40 cm from point B at right, real
 (4) 70 cm from point B at left, virtual

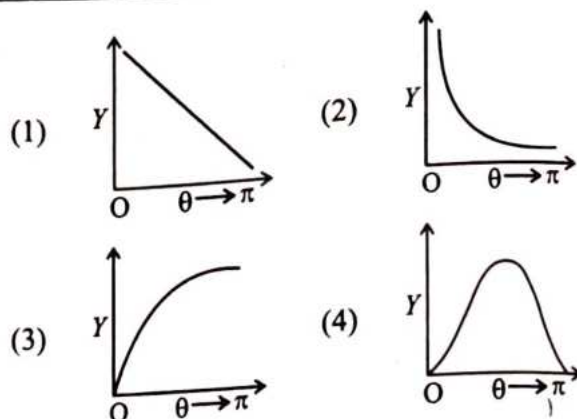
17. The work functions of aluminium and Gold are 4.1 eV and 5.1 eV respectively. The ratio of the slope of the stopping potential versus frequency plot for gold to that of aluminium is:

- (1) 1.24 (2) 1.5
 (3) 1 (4) 2

18. The logic operations performed by the given digital circuit is equivalent to:

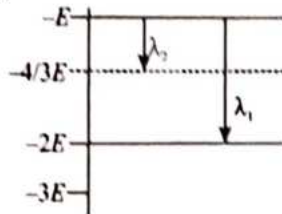


- (1) NAND (2) OR
(3) NOR (4) AND
19. A concave mirror for face viewing has focal length of 0.4 m. The distance at which you hold the mirror from your face in order to see your image upright with a magnification of 5 is:
(1) 0.16 m (2) 1.60 m
(3) 0.32 m (4) 0.24 m
20. An electron acceleration through a potential difference V_1 has a de-Broglie wavelength of λ . When the potential is changed to V_2 , its de-Broglie wavelength increases by 50%. The value of $\left(\frac{V_1}{V_2}\right)$ is equal to:
(1) 4 (2) $\frac{9}{4}$
(3) 3 (4) $\frac{3}{2}$
21. The energy of He^+ ion in its first excited state is, (The ground state energy for the hydrogen atom is -13.6 eV)
(1) -27.2 eV (2) -3.4 eV
(3) -54.4 eV (4) -13.6 eV
22. The graph which depicts the results of Rutherford gold foil experiment with α -particle is:
 θ : Scattering angle
 Y : Number of scattered α -particle detected (plots are schematic and not to scale)



23. A small particle of mass m moves in such a way that its potential energy $U = \frac{1}{2} m \omega^2 r^2$, where ω is constant and r is the distance of the particle from origin. Assuming Bohr's quantization of momentum and circular orbit, the radius of n th orbit will be:
(1) $\frac{1}{n}$ (2) n
(3) $\sqrt{n}$ (4) n^2
24. Speed of an electron in Bohr's 7th orbit for H-atom is $3.6 \times 10^6 \text{ m/s}$. The corresponding speed of the electron in 3rd orbit, (in m/s) is:
(1) 3.6×10^6 (2) 1.8×10^6
(3) 8.4×10^6 (4) 7.5×10^6
25. The ratio of intensities at two points P and Q on the screen in a Young's double slit experiment where phase difference between two waves of same amplitude are $\frac{\pi}{3}$ and $\frac{\pi}{2}$, respectively are:
(1) 2 : 3 (2) 3 : 1
(3) 3 : 2 (4) 1 : 3
26. The electron in a hydrogen atom first jumps from the third excited state to the second excited state and subsequently to the first excited state. The ratio of the respective wavelengths $\frac{\lambda_1}{\lambda_2}$ of the photons emitted in this process is:
(1) $\frac{20}{7}$ (2) $\frac{27}{5}$
(3) $\frac{7}{5}$ (4) $\frac{9}{7}$

27. Some energy levels of a molecule are shown in the figure. The ratio of the wavelengths $r = \frac{\lambda_1}{\lambda_2}$ is given by:



- (1) $r = \frac{2}{3}$ (2) $r = \frac{3}{4}$
(3) $r = \frac{1}{3}$ (4) $r = \frac{4}{3}$

28. Given below are two statements, one is labelled as Assertion (A) and the other is labelled as Reason (R).

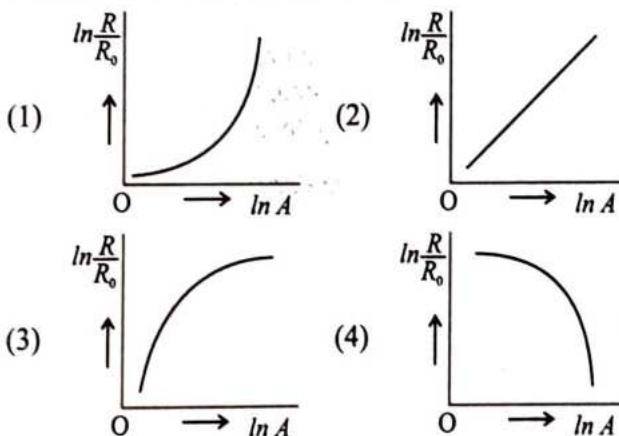
Assertion (A): The binding energy per nucleon is practically independent of the atomic number for nuclei of mass number in the range 30 to 170.

Reason (R): Nuclear force is short ranged.

In the light of the above statements, choose the most appropriate answer from the options given below

- (1) Both A and R are true and R is the correct explanation of A.
(2) Both A and R are true but R is not the correct explanation of A.
(3) A is true but R is false.
(4) A is false but R is true.

29. Which of the following figures represents the variation of $\left(\frac{R}{R_0}\right)$ with $\ln A$ (If R = radius of a nucleus and A = its mass number?)



30. A nucleus disintegrates into two nuclear parts which have their velocities in the ratio 2 : 1. The ratio of their nuclear sizes will be:

- (1) $2^{\frac{1}{3}} : 1$ (2) $1 : 3^{\frac{1}{3}}$
(3) $3^{\frac{1}{2}} : 1$ (4) $1 : 2^{\frac{1}{2}}$

31. If M_0 is the mass of an oxygen isotope ${}^{17}_8\text{O}$, M_p and M_n are the masses of a proton and a neutron respectively, the nuclear binding energy of the isotope is:

- (1) $(M_0 - 8M_p)c^2$
(2) $(8M_p + 9M_n - M_0)c^2$
(3) M_0c^2
(4) $(M_0 - 17M_n)c^2$

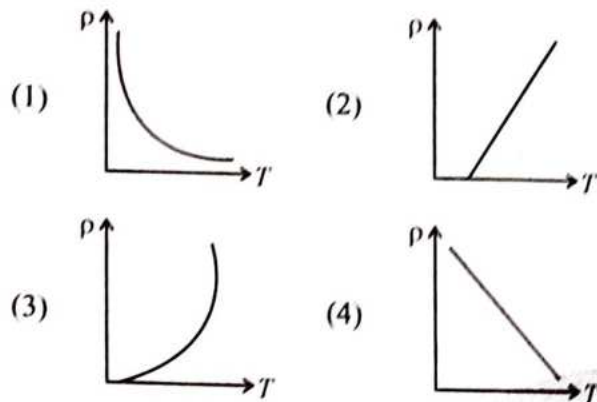
32. Match List-I (Electromagnetic wave type) with List-II (Its association/application) and select the correct option from the choices given below the lists.

List-I	List-II
(A) Infrared waves	(I) To treat muscular stain
(B) Radio waves	(II) For broadcasting
(C) X-ray	(III) To detect fracture of bones
(D) Ultraviolet	(IV) Absorbed by the ozone layer of the atmosphere

Choose the correct answer from the options given below:

- (1) A-IV, B-III, C-II, D-I
(2) A-I, B-II, C-IV, D-III
(3) A-III, B-II, C-I, D-IV
(4) A-I, B-II, C-III, D-IV

33. The resistivity (ρ) of semiconductor varies with temperature. Which of the following curve represents the **correct** behaviour?



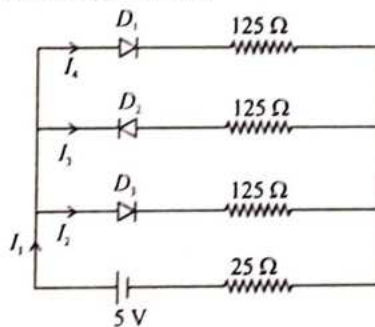
34. Given below are two statements: One is labelled as Assertion A and the other is labelled as Reason R:

Assertion (A): Diffusion current in a $p-n$ junction is greater than the drift current in magnitude if the junction is forward biased.

Reason (R): Diffusion current in a $p-n$ junction is from the n -side to the p -side, if the junction is forward biased.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Both A and R are correct and R is the correct explanation of A.
 - (2) A is not correct but R is correct.
 - (3) A is correct but R is not correct.
 - (4) Both A and R are correct but R is not the correct explanation of A.
35. If each diode has a forward bias resistance of 25Ω in the below circuit.



Which of the following options is **correct**.

- (1) $\frac{I_1}{I_2} = 1$
- (2) $\frac{I_2}{I_1} = 1$
- (3) $\frac{I_3}{I_4} = 1$
- (4) $\frac{I_1}{I_2} = 2$

Physics : Section-B (Q. No. 36 to 50)

36. n polarising sheets are arranged such that each makes an angle 45° with the preceding sheet. An unpolarised light of intensity I is incident into this arrangement. The output intensity is found to be

$\frac{I}{64}$. The value of n will be:

- (1) 3
- (2) 6
- (3) 4
- (4) 5

37. The angle of incidence at which reflected light is totally polarised for reflection from air to glass (refractive index n), is:

- (1) $\sin^{-1}(n)$
- (2) $\sin^{-1}\left(\frac{1}{n}\right)$
- (3) $\tan^{-1}\left(\frac{1}{n}\right)$
- (4) $\tan^{-1}(n)$

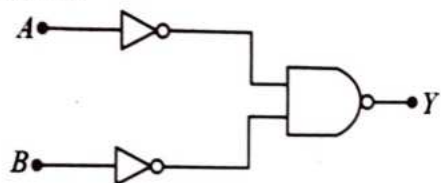
38. In a n -type semiconductor, which of the following statement is **true**?

- (1) Holes are majority carrier and trivalent atoms are dopants
- (2) Electron are majority carrier and trivalent atoms are dopants
- (3) Electrons are minority carries and pentavalent atoms are dopants
- (4) Holes are minority carries and pentavalent atoms are dopants

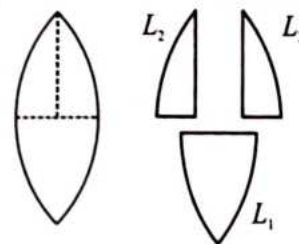
39. A plano-convex lens fits exactly into a planoconcave lens. Their plane surfaces are parallel to each other. If lenses are made of different materials of refractive indices μ_1 and μ_2 and R is the radius of curvature of the curved surface of the lenses, then the focal length of the combination is:

- (1) $\frac{2R}{(\mu_2 - \mu_1)}$
- (2) $\frac{R}{2(\mu_1 + \mu_2)}$
- (3) $\frac{R}{2(\mu_1 - \mu_2)}$
- (4) $\frac{R}{(\mu_1 - \mu_2)}$

40. The logic gate equivalent to the given logic circuit is:



- (1) NOR (2) NAND
(3) OR (4) AND
41. In Young's double slit experiment, 16 fringes are observed in a certain segment of the screen when light of wavelength 700 nm is used. If the wavelength of light is changed to 400 nm, the number of fringes observed in the same segment of the screen would be:
- (1) 24 (2) 30
(3) 18 (4) 28
42. A convex lens 'A' of focal length 20 cm and a concave lens 'B' of focal length 5 cm are kept along the same axis with a distance 'd' between them. If a parallel beam of light falling on 'A' leaves 'B' as a parallel beam, then the distance 'd' in cm will be:
- (1) 15 (2) 50
(3) 30 (4) 25
43. Two coherent sources of light interfere. The intensity ratio of sources is 1 : 4. For this interference pattern if the value of $\frac{I_{\max} + I_{\min}}{I_{\max} - I_{\min}}$ is equal to $\frac{2\alpha + 1}{\beta + 3}$ than $\frac{\alpha}{\beta}$ will be:
- (1) 1.5 (2) 2
(3) 0.5 (4) 1
44. A convex lens has power P . It is cut into two halves along its principal axis. Further one piece (out of the two halves) is cut into two halves perpendicular to the principal axis (as shown in figure). Choose the **incorrect** option for the reported pieces.



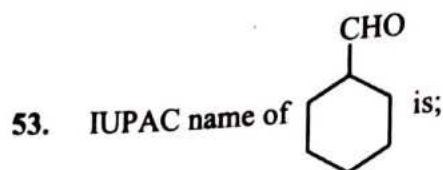
- (1) Power of $L_1 = \frac{P}{2}$
(2) Power of $L_2 = \frac{P}{2}$
(3) Power of $L_3 = \frac{P}{2}$
(4) Power of $L_1 = P$
45. According to Einstein's photoelectric equation, the graph between the kinetic energy of photoelectrons ejected and the frequency of incident radiation is:
- (1)
- (2)
- (3)
- (4)
46. Light of frequency 1.5 times the threshold frequency is incident on a photosensitive material. What will be the photoelectric current if the frequency is halved and intensity is doubled?
- (1) Four times (2) One-fourth
(3) Zero (4) Doubled
47. The ratio of kinetic energy to the total energy of an electron in a Bohr orbit of the hydrogen atom, is:
- (1) 2 : -1 (2) 1 : -1
(3) 1 : 1 (4) 1 : -2

48. If the nuclear force between two protons, two neutrons and between proton and neutron is denoted by F_{pp} , F_{nn} and F_{pn} respectively, then
- $F_{pp} \approx F_{nn} \approx F_{pn}$
 - $F_{pp} \neq F_{nn}$ and $F_{pp} = F_{nn}$
 - $F_{pp} = F_{nn} = F_{pn}$
 - $F_{pp} \neq F_{nn} \neq F_{pn}$
49. Consider a light travelling in air is incident into a medium of refractive index $\sqrt{2}n$. The incident angle is twice that of refracting angle. Then, the angle of incidence will be:
- $\sin^{-1}(\sqrt{n})$
 - $\cos^{-1}\left(\sqrt{\frac{n}{2}}\right)$
 - $\sin^{-1}(\sqrt{2}n)$
 - $2\cos^{-1}\left(\sqrt{\frac{n}{2}}\right)$
50. Two slits are made one millimetre apart and the screen is placed one metre away. What is the fringe separation when blue-green light of wavelength 500 nm is used?
- 0.5mm
 - 0.15mm
 - 0.8mm
 - 0.9mm

Chemistry : Section-A (Q. No. 51 to 85)

51. Correct statement is/are
- In alcohols, the oxygen of the -OH group is attached to carbon by a sigma (σ) bond formed by the overlap of a sp^3 hybridised orbital of carbon with a sp^3 hybridised orbital of oxygen.
 - The bond angle in alcohols is slightly less than the tetrahedral angle ($109^\circ 28'$).
 - In phenols, the -OH group is attached to sp^2 hybridised carbon of an aromatic ring.
 - The carbon-oxygen bond length (136 pm) in phenol is more than that in methanol.
- (a), (b) and (c)
 - (b), (c) and (d)
 - (a) and (d)
 - (b) and (c)

52. Which one is a primary alcoholic group?
- $-\text{CH}_2\text{OH}$
 - >CHOH
 - >C-OH
 - $\begin{array}{c} \text{>C}-\text{C}< \\ | \quad | \\ \text{OH} \quad \text{OH} \end{array}$



- Cyclohexanal
- Cyclohexane carbaldehyde
- Cyclohexyl carbaldehyde
- None of these

54. Which one of the following is a secondary amine?
- 2-Butanamine
 - N-Methylpiperidine
 - N-Methyl-2-pentanamine
 - p-Anisidine

55. Gluconic acid is not obtained in;

- Glucose $\xrightarrow{\text{Br}_2, \text{H}_2\text{O}}$
- Glucose $\xrightarrow[\text{(ii) HNO}_3]{\text{Tollen's Reagent}}$
- Glucose $\xrightarrow[\text{(ii) H}_3\text{O}^+]{\text{(i) Fehling's solution}}$
- Glucose $\xrightarrow{\text{HNO}_3}$

56. Carbonate salts evolve which gas on treating with dilute sulphuric acid;

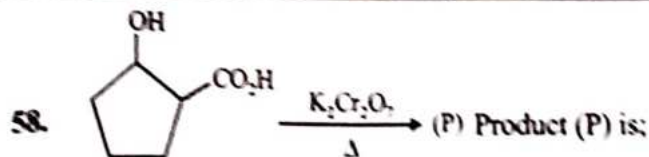
- SO_2
- NO_2
- Cl_2
- CO_2

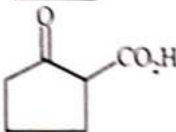
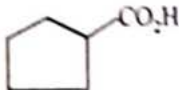
57. Match List-I with List-II

List-I (Compounds)	List-II (Classification)
A. CH_3CHCl_2	I. Vinyl halide
B. $\text{CH}_2\text{ClCH}_2\text{Cl}$	II. Alkylidene halide
C. $\text{CHCl}=\text{CH}_2$	III. Alkylene dihalide
D. $\text{ClCH}_2-\text{CH}=\text{CH}_2$	IV. Allyl halide

Choose the correct answer from the options given below:

- A-III, B-I, C-II, D-IV
- A-II, B-III, C-I, D-IV
- A-II, B-I, C-III, D-IV
- A-II, B-III, C-IV, D-I

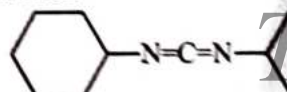
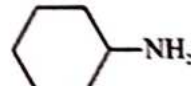
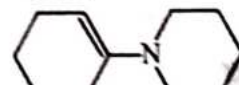
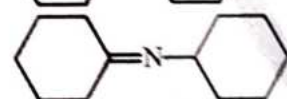


- (1)  (2) 
 (3)  (4) 

59. IUPAC name of CCl_3CHO is:

- (1) Chloral
 (2) Trichloro acetaldehyde
 (3) 1,1,1-Trichloroethanal
 (4) 2,2,2-Trichloroethanal

60. Which of the following compound is an enamine?

- (1) 
 (2) 
 (3) 
 (4) 

61. Invert sugar is an equimolar mixture of;

- (1) D-Glucose and D-Fructose
 (2) D-Glucose and L-Fructose
 (3) D-Glucose and L-Glucose
 (4) D-Fructose and L-Fructose

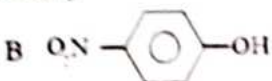
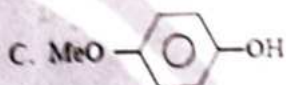
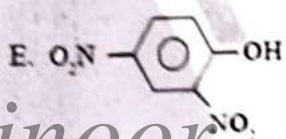
62. For the confirmatory test of sulphide ion a piece of filter paper dipped in lead acetate solution turns black on exposure to the (H_2S) gas due to the formation of;

- (1) Lead Carbonate
 (2) Lead Nitrate
 (3) Lead Acetate
 (4) Lead Sulphide

63. The ease with which an alcohol can be halogenated by treating it with HX in the presence of anhydrous $ZnCl_2$ is;

- (1) Tertiary alcohol < secondary alcohol < primary alcohol
 (2) Tertiary alcohol > secondary alcohol > primary alcohol
 (3) Tertiary alcohol > secondary alcohol < primary alcohol
 (4) Tertiary alcohol < secondary alcohol > primary alcohol

64. Select the correct order of acidity

- A. $Bu-OH$ B. 
 C.  D. 
 E. 

I. (B) > (E) and C = D

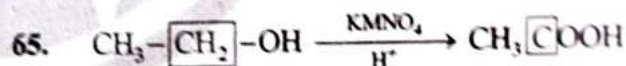
II. (D) < (C)

III. (A) = (B) < (C)

IV. (B) > (C) and (E) > (D)

Choose the correct answer from the options given below:

- (1) (II) only (2) (II) and (III)
 (3) (III) and (IV) (4) (IV) only



Hybridisation of carbon change as;

- (1) $sp-sp^2$ (2) sp^2-sp
 (3) sp^3-sp^2 (4) sp^3-sp

66. Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**
Assertion (A): The main product of reaction of alcoholic silver nitrite and ethyl bromide is nitroethane.

Reason (R): Silver nitrite is predominantly covalent compound.

In the light of the above statements, choose the correct answer from the options given below:

- (1) A is false but R is true.
- (2) A is true but R is false.
- (3) Both A and R are true and R is the correct explanation of A.
- (4) Both A and R are true but R is NOT the correct explanation of A.

67. Match List-I with List-II

List-I (Information)	List-II (Amino acids)
A. Alcohol containing amino acid	I. Valine
B. Basic amino acid	II. Serine
C. Essential amino acid	III. Glycine
D. Optically inactive amino acid	IV. Asparagine

Choose the correct answer from the options given below:

- (1) A-III, B-I, C-II, D-IV
- (2) A-II, B-IV, C-I, D-III
- (3) A-II, B-I, C-III, D-IV
- (4) A-II, B-III, C-IV, D-I

68. The preliminary test of nitrite ion is performed by;

- (1) Conc. H_2SO_4
- (2) Dil. H_2SO_4
- (3) Conc. HNO_3
- (4) Dil. HNO_3

69. Which of the following statement(s) is/are correct?

- A. The boiling points of isomeric haloalkanes decrease with increase in branching.
- B. Methyl chloride, methyl bromide, ethyl chloride are gases at room temperature.
- C. Density of n-propylchloride is less than n-propylbromide.

- (1) A and B only
- (2) B and C only
- (3) A, and C only
- (4) A, B and C

70. Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**
Assertion (A): Alcohols behave as bases in the presence of mineral acids.

Reason (R): This is because of lone electron pair on the oxygen atom.

In the light of the above statements, choose the correct answer from the options given below:

- (1) A is false but R is true.
- (2) A is true but R is false.
- (3) Both A and R are true and R is the correct explanation of A.
- (4) Both A and R are true but R is NOT the correct explanation of A.

71. The gas evolved in the preliminary test of Chloride ion is;

- (1) SO_2
- (2) Cl_2
- (3) HCl
- (4) HN_3

72. Which of the following reagents can be used to convert primary amides into primary amines containing the -same number of carbon atoms?

- (1) $\text{Br}_2 + \text{NaOH}$
- (2) LiAlH_4
- (3) $\text{Sn} + \text{HCl}$
- (4) $\text{Na} + \text{C}_2\text{H}_5\text{OH}$

73. Which of the following is not a pyrimidine base?

- (1) Uracil
- (2) Guanine
- (3) Cytosine
- (4) Thymine

74. The formula of Mohr's salt is;

- (1) $\text{Fe}(\text{SO}_4) \cdot 6\text{H}_2\text{O}$
- (2) $(\text{NH}_4)_2\text{SO}_4 \cdot 6\text{H}_2\text{O}$
- (3) $(\text{NH}_4)_2\text{SO}_4 \cdot \text{Fe}(\text{SO}_4) \cdot 6\text{H}_2\text{O}$
- (4) $(\text{NH}_4)_2\text{SO}_4 \cdot \text{Fe}(\text{SO}_4) \cdot 10\text{H}_2\text{O}$

75. Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**

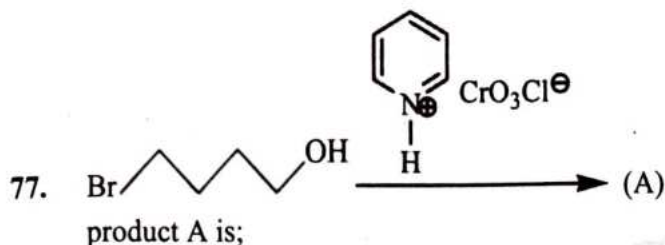
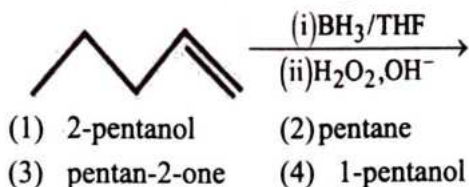
Assertion (A): Hydrolysis of (-)-2-Bromooctane proceeds with inversion of configuration.

Reason (R): This reaction proceeds through the formation of a carbocation.

In the light of the above statements, choose the correct answer from the options given below:

- (1) A is false but R is true.
- (2) A is true but R is false.
- (3) Both A and R are true and R is the correct explanation of A.
- (4) Both A and R are true but R is NOT the correct explanation of A.

76. The product of the following reaction is;



- (1)
- (2)
- (3)
- (4)

78. Match List-I with List-II

List-I	List-II
A. Hinsberg's reagent	I. $\text{Br}_2 + \text{NaOH}$
B. Reagent used in isocyanide test	II. $\text{C}_6\text{H}_5\text{SO}_2\text{Cl}$
C. Reagent used in Hoffmann bromamide degradation reaction	III. $\text{CHCl}_3 + \text{KOH}$
D. Reagent used to convert nitrobenzene to aniline	IV. $\text{Sn} + \text{HCl}$

Choose the **correct** answer from the options given below:

- (1) A-III, B-I, C-II, D-IV
(2) A-II, B-III, C-I, D-IV
(3) A-II, B-I, C-III, D-IV
(4) A-II, B-III, C-IV, D-I

79. Given below are two statements:

Statement-I: Glucose and mannose are Epimers.

Statement-II: Glucose and fructose are Functional isomers.

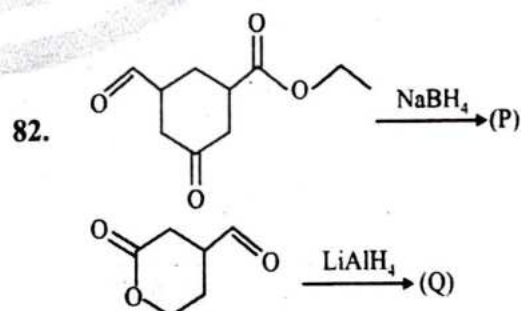
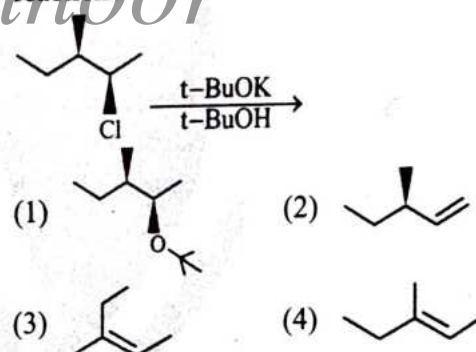
In the light of the above statements, choose the **correct** answer from the options given below:

- (1) Statement I is correct but Statement II is incorrect.
(2) Statement I is incorrect but Statement II is correct.
(3) Both Statement I and Statement II are correct.
(4) Both Statement I and Statement II are incorrect.

80. Which compound turn lime water milky;

- (1) CO_2 (2) NO_2
(3) Cl_2 (4) All of these

81. What is the major product of the following reaction?



Sum of number of 1° alcoholic groups in product (P) and (Q) is;

- (1) 1 (2) 2
(3) 3 (4) 4

83. Given below are two statements:
Statement I: Only aromatic aldehydes undergo Cannizzaro reaction.

Statement II: Those aldehydes which have α -hydrogen atom undergo Cannizzaro reaction.
 In the light of the above statements, choose the correct answer from the options given below:

- (1) Statement I is correct but Statement II is incorrect.
- (2) Statement I is incorrect but Statement II is correct.
- (3) Both Statement I and Statement II are correct.
- (4) Both Statement I and Statement II are incorrect.

84. Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**

Assertion (A): Carbylamine reaction involves chemical reaction between 1° amine and chloroform in basic medium.

Reason (R): In carbylamine reaction, $-\text{NH}_2$ group changes in to $-\text{CN}$ group.

In the light of the above statements, choose the correct answer from the options given below:

- (1) A is false but R is true.
- (2) A is true but R is false.
- (3) Both A and R are true and R is the correct explanation of A.
- (4) Both A and R are true but R is NOT the correct explanation of A.

85. Scurvy is caused by the deficiency of;

- (1) Vitamin B₁
- (2) Vitamin C
- (3) Vitamin K
- (4) Vitamin A

Chemistry : Section-B (Q. No. 86 to 100)

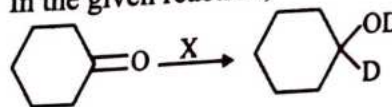
86. When a salt is warm with dilute H_2SO_4 it gives a gas which smells like rotten eggs, the salt is;

- (1) Sulphate Salt
- (2) Carbonate Salt
- (3) Sulphide Salt
- (4) Chloride Salt

87. The correct order of reactivities of the following alkyl halides for a $\text{S}_{\text{N}}2$ reaction is;

- (1) $\text{RF} > \text{RCl} > \text{RBr} > \text{RI}$
- (2) $\text{RF} > \text{RBr} > \text{RCl} > \text{RI}$
- (3) $\text{RCl} > \text{RBr} > \text{RF} > \text{RI}$
- (4) $\text{RI} > \text{RBr} > \text{RCl} > \text{RF}$

88. In the given reaction,

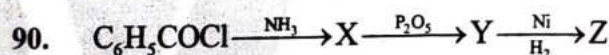


[X] will be;

- (1) $\text{LiAlD}_4/\text{D}_2\text{O}$
- (2) $\text{NaBD}_4/\text{H}_2\text{O}$
- (3) $\text{LiAlD}_4/\text{HOH}$
- (4) $\text{NaBH}_4/\text{D}_2\text{O}$

89. o-xylene when oxidised in presence of V_2O_5 the product is;

- (1) Benzoic acid
- (2) Phenyl acetic acid
- (3) Phthalic acid
- (4) Acetic acid



The end product in the above sequence of reactions is;

- (1) benzoic acid
- (2) aniline
- (3) benzyl amine
- (4) benzonitrile

91. Assume that a particular amino acid has an isoelectric point of 6.0. In a solution at pH 1.0, which of the following species will predominate?

- (1) $\text{H}_3\text{N}^+\text{CH(R)CO}_2\text{H}$
- (2) $\text{H}_2\text{NCH(R)CO}_2\text{H}$
- (3) $\text{H}_3\text{N}^+\text{CH(R)CO}_2^-$
- (4) $\text{H}_2\text{NCH(R)CO}_2^-$

92. A salt containing Zn^{2+} after heating its colour becomes;

- (1) Blue
- (2) Yellow
- (3) White
- (4) Red

93. Chlorobenzene is formed by reaction of chlorine with benzene in the presence of AlCl_3 . Which of the following species attacks the benzene ring in this reaction?

- (1) Cl^- (2) Cl^+
(3) AlCl_3 (4) $[\text{AlCl}_4]^-$

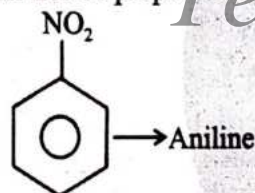
94. Which of the following is **not** a method for preparation of phenol?

- (1) Dows process
(2) From diazonium salt
(3) By decarboxylation of salicylic acid
(4) By the decarboxylation of sodium benzoate

95. Schiff's base is prepared from;

- (1) Carbonyl compound and primary amine
(2) Carbonyl compound and secondary amine
(3) Carbonyl compound and tertiary amine
(4) All of these

96. Aniline will **not** be prepared in presence of;



- (1) $\text{Sn} + \text{HCl}$ (2) $\text{Fe} + \text{HCl}$
(3) $\text{Zn} + \text{NH}_4\text{Cl}$ (4) H_2/Pd

97. The hormone which controls the processes of burning of fats protein, and carbohydrates and liberates energy in the body is:

- (1) Thyroxine (2) Adrenaline
(3) Insulin (4) Cortisone

98. Which of the following is not a group V cation?

- (1) Ba^{2+} (2) Ca^{2+}
(3) Ni^{2+} (4) Sr^{2+}

99. If chloroform is left open in air in the presence of sunlight, it gives;

- (1) Carbon tetrachloride
(2) Carbonyl chloride
(3) Mustard gas
(4) Chlorine gas

100. p-Hydroxyazobenzene is;

- (1) an orange dye (2) a yellow dye
(3) a red dye (4) an orange-red dye.

Botany : Section-A (Q. No. 101 to 135)

101. The salt concentration [measured as salinity in parts per thousand (ppt)], is ___A___ in inland waters, ___B___ in the sea and ___C___ in some hypersaline lagoons.

- (1) $A \rightarrow$ more than 15, $B \rightarrow$ 30-35, $C \rightarrow$ more than 100
(2) $A \rightarrow$ 30-35, $B \rightarrow$ more than 100, $C \rightarrow$ less than 5
(3) $A \rightarrow$ less than 5, $B \rightarrow$ more than 100, $C \rightarrow$ 30-35
(4) $A \rightarrow$ less than 5, $B \rightarrow$ 30-35, $C \rightarrow$ more than 100

102. Given below are two statements:

Statement I: The physico-chemical (abiotic) components alone characterise the habitat of an organism completely.

Statement II: Diversity in the resources it utilises and a distinct functional role in the ecological system, are some of the criteria that comprise niche of an organism.

In the light of the above statements, choose the *most appropriate* answer from the options given below:

- (1) Statement I is correct but Statement II is incorrect.
(2) Statement I is incorrect but Statement II is correct.
(3) Both Statement I and Statement II are correct.
(4) Both Statement I and Statement II are incorrect.

103. Bottled fruit juices bought from the market are clearer as compared to those made at home. This is because the bottled juices are clarified by the use of:

- (1) pectinases and lipases.
(2) lipases and proteases.
(3) lipases and streptokinase.
(4) pectinases and proteases.

104. Choose the **incorrect** statement w.r.t. light as a major abiotic factor.

- (1) Many species of tall trees growing in forests are adapted to photosynthesise optimally under very low light conditions.
- (2) Many plants are dependent on sunlight to meet their photoperiodic requirement for flowering.
- (3) For many animals too, light is important as cues for timing their foraging, reproductive and migratory activities.
- (4) The availability of light on land is closely linked with that of temperature since the sun is the source for both.

105. The efficiency of feeding in one species might be reduced due to the inhibitory presence of the other species, even if the food and space are abundant is called:

- (1) competitive exclusion.
- (2) competitive release.
- (3) interference competition
- (4) resource partitioning.

106. Arrange the following steps in the correct sequence as they occur in the sewage treatment process.

- (a) Effluent is passed into large aeration tanks that allows vigorous growth of aerobic microbes.
- (b) Floating debris is removed through sequential filtration.
- (c) A small part of activated sludge is pumped back into the aeration tank as inoculum and the rest goes to anaerobic digesters.
- (d) Sedimentation allows grit and small pebbles to settle out.
- (e) Bacterial 'flocs' sediment in a settling tank after significant BOD reduction.
- (f) Methane, hydrogen sulphide, and carbon dioxide are produced in anaerobic digesters.
- (g) Treated effluent is released into natural water bodies.

Choose the correct sequence from the options given below:

- (1) (b), (d), (a), (e), (c), (f), (g)
- (2) (a), (b), (c), (d), (e), (f), (g)
- (3) (d), (b), (a), (e), (f), (c), (g)
- (4) (b), (a), (d), (e), (c), (f), (g)

107. Given below are two statements:

Statement I: Primary production is defined as the amount of biomass or organic matter produced per unit area over a time period by plants during photosynthesis.

Statement II: Primary production is expressed in terms of $\text{gm}^{-2} \text{yr}^{-1}$ or $(\text{kcal m}^{-2}) \text{yr}^{-1}$.

In the light of the above statements, choose the *most appropriate* answer from the options given below:

- (1) Statement I is correct but Statement II is incorrect.
- (2) Statement I is incorrect but Statement II is correct.
- (3) Both Statement I and Statement II are correct.
- (4) Both Statement I and Statement II are incorrect.

108. Select the **incorrect** match w.r.t sacred groves found in different states.

- (1) Khasi and Jaintia Hills - Meghalaya
- (2) Aravalli Hills - Rajasthan
- (3) Western Ghat region - Karnataka
- (4) Sarguja, Chanda and Bastar - Gujarat

109. The bioactive molecule cyclosporin A is used for:

- (1) dissolving blood clots.
- (2) immunosuppression.
- (3) lowering blood pressure.
- (4) lowering cholesterol level.

110. Given below are two statements:

Statement I: Zoological parks, botanical gardens and wildlife safari parks serve the purpose of *ex situ* conservation of threatened animals and plants.

Statement II: In recent years *ex situ* conservation has advanced beyond keeping threatened species in enclosures.

In the light of the above statements, choose the *most appropriate* answer from the options given below:

- (1) Statement I is correct but Statement II is incorrect.
- (2) Statement I is incorrect but Statement II is correct.
- (3) Both Statement I and Statement II are correct.
- (4) Both Statement I and Statement II are incorrect.

111. Cyclosporin A and statins are, respectively, produced by;
- (1) *Trichoderma polysporum* and *Monascus purpureus*.
 - (2) *Aspergillus niger* and *Monascus purpureus*.
 - (3) *Aspergillus niger* and *Saccharomyces cerevisiae*.
 - (4) *Trichoderma polysporum* and *Saccharomyces cerevisiae*.

112. Identify the **incorrect** statement.

- (1) All animals depend on plants (directly or indirectly) for their food needs. They are hence called consumers and also heterotrophs.
- (2) If animals feed on the producers, the plants, they are called primary consumers.
- (3) If the animals eat other animals which in turn eat the plants (or their produce) they are called secondary consumers.
- (4) The secondary consumers will always be herbivores.

113. Given below are two statement: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: No energy that is trapped into an organism remains in it for ever.

Reason R: The energy trapped by the producer, is either passed on to a consumer or the organism dies.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) A is true but R is false.
- (2) A is false but R is true.
- (3) Both A and R are true and R is the correct explanation of A.
- (4) Both A and R are true but R is NOT the correct explanation of A.

114. Match the List-I with List-II.

List-I	List-II
A. Genetic diversity	I. Combined diversity at all the levels of biological organisation
B. Biodiversity	II. <i>Rauwolfia vomitoria</i> in Himalayan ranges
C. Species diversity	III. India has more ecosystem diversity than Norway
D. Ecological diversity	IV. Diversity of amphibians is greater in Western Ghats than Eastern Ghats

Choose the **correct** answer from the options given below:

- (1) A-II, B-I, C-IV, D-III
- (2) A-I, B-II, C-III, D-IV
- (3) A-II, B-III, C-I, D-IV
- (4) A-III, B-II, C-I, D-IV

115. Which of the following are considered direct economic benefits of biodiversity ?

- (a) Medicinal plants
- (b) Construction materials
- (c) Aesthetic pleasures
- (d) Food resources

Choose the correct answer from the options given below:

- (1) (a), (b) and (d) only
- (2) (a), (c) and (d) only
- (3) (b), (c) and (d) only
- (4) (a), (b), (c) and (d)

116. The components of the ecosystem are seen to function as a unit when following aspects are considered **except**:

- (1) decomposition. (2) stratification.
- (3) energy flow. (4) nutrient cycling.

117. Among the vertebrates, maximum number of species is observed for ____.

- (1) fishes (2) reptiles
- (3) birds (4) mammals

118. Choose the **odd** one out w.r.t. adaptations in the organisms.

- (1) Biochemical adaptation are seen in organisms living in great depth in the ocean to face crushing pressure.
- (2) Allen's rule is seen in mammals living in colder climates
- (3) Altitude sickness is caused because of body not getting enough oxygen due to low atmospheric pressure at high altitude.
- (4) Desert lizards lack the physiological and behavioural means to manage to their body temperature

119. Given below are two statements:

Statement I: The majority of baculoviruses used as biological control agents are in the genus *Nucleopolyhedrovirus*.

Statement II: *Nucleopolyhedrovirus* are excellent candidates for species-specific, narrow spectrum insecticidal applications.

In the light of the above statements, choose the **most appropriate** answer from the options given below:

- (1) Statement I is correct but Statement II is incorrect.
- (2) Statement I is incorrect but Statement II is correct.
- (3) Both Statement I and Statement II are correct.
- (4) Both Statement I and Statement II are incorrect.

120. Given below are two statement: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion (A): The Monarch butterfly is highly distasteful to its predator (bird) because of a special chemical present in its body.

Reason (R): Some prey species are poisonous and therefore avoided by the predators.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) A is true but R is false.
- (2) A is false but R is true.
- (3) Both A and R are true and R is the correct explanation of A.
- (4) Both A and R are true but R is NOT the correct explanation of A.

121. All of the following are **correct** w.r.t. mycorrhiza, **except**:

- (1) helps plants in absorption of phosphorus from the soil.
- (2) provides resistance to root-borne pathogens.
- (3) mycorrhiza is a parasitic association between fungi and plants.
- (4) provides tolerance to salinity and drought in plants.

122. Match List-I with List-II.

List-I		List-II	
(A) Habitat loss	(I)	Extinction of Steller's sea cow	
(B) Over-exploitation	(II)	Invasive growth of <i>Eichhornia</i>	
(C) Alien species invasion	(III)	Extinction of a parasite with the extinction of host	
(D) Co-extinction	(IV)	Cutting forest for growing soya beans	

Choose the **correct** answer from the options given below:

	A	B	C	D
(1)	I	II	III	IV
(2)	III	I	IV	II
(3)	IV	III	II	I
(4)	IV	I	II	III

123. Mark **true** (T) and **false** (F) for the following statements and choose the **correct** option.

A. With increasing resources put into 'bioprospecting' (exploring molecular, genetic and species-level diversity for products of economic importance), nations endowed with rich biodiversity can expect to reap enormous benefits.

B. Amazon forest is estimated to produce, through photosynthesis, 20 per cent of the total oxygen in the earth's atmosphere.

	A	B
(1)	T	T
(2)	T	F
(3)	F	F
(4)	F	T

124. Identify the **incorrect** statement(s) from the following.

- A. In *in situ* approach of conservation of biodiversity, threatened animals and plants are taken out from their natural habitat and placed in special setting where they can be protected and given special care.
 - B. Zoological parks, botanical gardens and wildlife safari parks serve the purpose of *ex situ* conservation.
 - C. In recent years *in situ* conservation has advanced beyond keeping threatened species in enclosures.
- (1) C only (2) A, B and C
(3) A and C only (4) A only

125. Read the following statements.

- A. 'Rivet popper hypothesis' is given by Stanford ecologist Paul Ehrlich to explain importance of species richness for the safety of an ecosystem.
- B. According to this hypothesis, rivets on the wings of an airplane is compared with the key species that drive major ecosystem functions.

Choose the *most appropriate* answer from the options given below:

- (1) A is correct, B is incorrect.
- (2) A is incorrect, B is correct.
- (3) A and B are correct.
- (4) A and B are incorrect.

126. Choose the **correct** option w.r.t. the products which are produced by the usage of yeast:

- (1) citric acid and lactic acid.
- (2) lipase and pectinase.
- (3) cheese and butter.
- (4) bread and beer.

127. Select the **incorrect** statement w.r.t. *Calotropis*.

- (1) It is a poisonous weed.
- (2) It contains toxic chemicals i.e. cardiac glycosides.
- (3) It develops morphological means to avoid predation.
- (4) Cattles and goats never browse on this plant.

128. Given below are two statement: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion (A): When resources are limited, the competitively superior species will eventually eliminate the other species.

Reason (R): In general, herbivores and plants appear to be more adversely affected by competition than carnivores.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) A is true but R is false.
- (2) A is false but R is true.
- (3) Both A and R are true and R is the correct explanation of A.
- (4) Both A and R are true but R is NOT the correct explanation of A.

129. Sequence the given below list of biomes from lowest to highest mean annual precipitation.

- (a) Desert (b) Grassland
- (c) Tropical forest (d) Coniferous forest

Choose the **correct** sequence from the options given below:

- (1) (b), (a), (d), (c) (2) (d), (b), (a), (c)
- (3) (a), (b), (d), (c) (4) (d), (b), (c), (a)

130. Given below are two statements:

Statement I: Majority of the organisms can tolerate and thrive in a wide range of temperatures and are called eurythermal.

Statement II: A few organisms which are restricted to a narrow range of temperatures are called stenothermal.

In the light of the above statements, choose the *most appropriate* answer from the options given below:

- (1) Statement I is correct but Statement II is incorrect.
- (2) Statement I is incorrect but Statement II is correct.
- (3) Both Statement I and Statement II are correct.
- (4) Both Statement I and Statement II are incorrect.

131. The predominant gas in biogas is:
 (1) hydrogen sulphide.
 (2) carbon dioxide.
 (3) methane.
 (4) ethane.

132. The solids which settle after a primary treatment of sewage are called:
 (1) primary sludge. (2) activated sludge.
 (3) flocs. (4) effluent.

133. Read the following statements and select the correct ones.

- A. Endoparasites live outside the host body.
 B. The evolutions of the flower and its pollinator species are tightly linked with one another.
 C. The wasp pollinates the fig inflorescence while searching for suitable egg-laying sites.

Choose the *most appropriate* answer from the options given below:

- (1) A and B only
 (2) A and C only
 (3) A only
 (4) B and C only

134. Match List-I with List-II:

List-I	List-II
(A) Regulator	(I) Siberian crane
(B) Conformers	(II) Fungi
(C) Migrators	(III) Shrews
(D) Suspend	(IV) Mammals

Choose the **correct** answer from the options given below:

- (1) A-IV, B-III, C-I, D-II
 (2) A-III, B-IV, C-II, D-I
 (3) A-I, B-II, C-III, D-IV
 (4) A-II, B-I, C-III, D-IV

135. How many of the following statements are correct w.r.t. *Trichoderma*?

- A. Developed for use in the treatment of plant diseases.
 B. It's species are free living fungi.
 C. They are very common in shoot ecosystems.

- D. They are effective biocontrol agents for several plant pathogens.

Choose the *most appropriate* answer from the options given below:

- (1) 4 (2) 3
 (3) 2 (4) 1

Botany : Section-B (Q. No. 136 to 150)

136. Match the List-I and List-II.

List-I (Biocontrol agent)	List-II (Pest or pathogen)
(A) Lady bird	(I) Arthropods/insects
(B) Dragonfly	(II) Mosquitoes
(C) <i>B. thuringiensis</i>	(III) Butterfly caterpillars
(D) Baculoviruses	(IV) Aphids

Choose the **correct** answer from the options given below:

- (1) A-IV, B-III, C-I, D-II
 (2) A-III, B-IV, C-II, D-I
 (3) A-I, B-III, C-II, D-IV
 (4) A-IV, B-II, C-III, D-I

137. Choose the **incorrectly** matched pair.

- (1) *Ophrys* and male bee - Competition
 (2) *Chathamalus* and *Balanus* - Competition
 (3) *Cuscuta* on hedge plants - Parasitism
 (4) *Pisaster* - Predator

138. Gross primary productivity minus ___A___ is the net ___B___ productivity.

Select the **correct** option.

- (1) A = respiration losses, B = secondary
 (2) A = producers, B = primary
 (3) A = respiration losses, B = primary
 (4) A = consumers, B = secondary

139. Among these which one is known as 'lungs of the planet'?

- (1) Western ghats of Maharashtra
 (2) Amazon rain forest
 (3) Aravalli Hills
 (4) Khasi and Jaintia Hills

140. Given below are two statements:

Statement I: Species richness does not contribute to the well-being of an ecosystem

Statement II: Increased diversity contributed to higher productivity in Tilman's experiments.

In the light of the above statements, choose the *most appropriate* answer from the options given below:

- (1) Statement I is correct but Statement II is incorrect.
- (2) Statement I is incorrect but Statement II is correct.
- (3) Both Statement I and Statement II are correct.
- (4) Both Statement I and Statement II are incorrect.

141. Match the List-I with List-II.

List-I (Microbe)	List-II (used in production of)
(A) <i>Propionibacterium sharmanii</i>	(I) Penicillin
(B) <i>Saccharomyces cerevisiae</i>	(II) Blood-cholesterol lowering agent
(C) <i>Penicillium notatum</i>	(III) Swiss cheese
(D) <i>Monascus purpureus</i>	(IV) Bread

Choose the **correct** answer from the options given below:

- | | A | B | C | D |
|-----|-----|-----|----|-----|
| (1) | I | III | II | IV |
| (2) | III | IV | I | II |
| (3) | III | II | I | IV |
| (4) | IV | I | II | III |

142. Match the List-I with List-II.

List-I	List-II
(A) $-,-$	(I) Amensalism
(B) $+,-$	(II) Commensalism
(C) $+,0$	(III) Parasitism
(D) $-,0$	(IV) Competition

Choose the **correct** answer from the options given below:

- (1) A-IV, B-III, C-I, D-II
- (2) A-IV, B-III, C-II, D-I
- (3) A-IV, B-I, C-II, D-III
- (4) A-I, B-III, C-II, D-IV

143. From the given examples, identify the ones that explains loss of biodiversity due to overexploitation of natural resources.

- A. Extinction of 200 species of cichlid fish due to introduction of Nile perch in Lake Victoria in east Africa.
- B. Threat caused to native species of crops by invasive weed like *Lantana*.
- C. Extinction of passenger pigeon.
- D. Endangerment of commercially important marine species due to over-harvesting of fishes.

Choose the *most appropriate* answer from the options given below:

- (1) C and D only
- (2) A and B only
- (3) A and C only
- (4) B and D only

144. Which of the following are critical factors for biodiversity conservation?

- (a) Habitat restoration
- (b) Exotic species removal
- (c) Water conservation
- (d) Genetic diversity maintenance
- (e) Pollution reduction

Options:

- (1) (a), (b) and (e) only
- (2) (a), (d) and (e) only
- (3) (a), (c) and (e) only
- (4) (a), (b), (d) and (e)

145. Which of the following interactions are examples of mutualism?

- (a) A lichen composed of a fungus and an alga
- (b) A tapeworm in the intestine of a human
- (c) A barnacle attached to a whale
- (d) An orchid growing on a tree branch

Choose the correct answer from the options given below:

- (1) (a) only
- (2) (a) and (b) only
- (3) (a), (c), and (d) only
- (4) (b), (c), and (d) only

146. Given below are two statement: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion A: The ethical argument for conserving biodiversity relates to what we owe to millions of plant, animal and microbe species with whom we share this planet.

Reason R: Philosophically or spiritually, we need to realise that every species has an intrinsic value, even if it may not be of current or any economic value to us.

In the light of the above statements, choose the correct answer from the options given below:

- (1) A is true but R is false.
- (2) A is false but R is true.
- (3) Both A and R are true and R is the correct explanation of A.
- (4) Both A and R are true but R is NOT the correct explanation of A.

147. Select the incorrect statement regarding decomposition.

- (1) Decomposition is largely an oxygen requiring process.
- (2) The rate of decomposition is controlled by chemical composition of detritus and climatic factors.
- (3) Decomposition rate is slower if detritus is rich in nitrogen and water-soluble substances like sugars and quicker, if detritus is rich in chitin and lignin.
- (4) Temperature and soil moisture are the most important climatic factors that regulate decomposition through their effects on the activities of soil microbes

148. Match List-I with List-II.

List-I (Compound)	List-II (Source)
(A) Citric acid	(I) <i>Aspergillus niger</i>
(B) Butyric acid	(II) <i>Clostridium butylicum</i>
(C) Acetic acid	(III) <i>Acetobacter aceti</i>
(D) Ethanol	(IV) <i>Saccharomyces cerevisiae</i>

Choose the correct answer from the options given below:

- | | | | |
|--------|-----|-----|-----|
| A | B | C | D |
| (1) IV | III | II | I |
| (2) I | II | IV | III |
| (3) IV | II | III | I |
| (4) I | II | III | IV |

149. The annual net primary productivity of the oceans is:

- (1) 170 billion tons.
- (2) 55 billion tons.
- (3) 115 billion tons.
- (4) 70 billion tons.

150. Select the incorrect statement.

- (1) The base of each ecological pyramid represents the producers or the first trophic level while the apex represents tertiary or top level consumer.
- (2) A given organism can occupy only one trophic level.
- (3) The trophic level represents a functional level, not a species as such.
- (4) Energy at a lower trophic level is always more than at a higher level.

Zoology : Section-A (Q. No. 151 to 185)

151. *Bacillus thuringiensis* forms protein crystals which contain a:

- (1) simple protein.
- (2) non-toxic insecticidal protein.
- (3) toxic insecticidal protein.
- (4) None of these

152. Match List-I with List-II:

List-I	List-II
(A) Bioreactors	(I) Separation of DNA fragments
(B) Gel electrophoresis	(II) Production of large quantities of products
(C) Polymerase chain reaction	(III) Detection of pathogen, based on antigen-antibody interaction
(D) ELISA	(IV) Amplification of DNA

Choose the correct answer from the options given below:

- (1) A-III, B-II, C-IV, D-I
- (2) A-II, B-I, C-IV, D-III
- (3) A-IV, B-III, C-II, D-I
- (4) A-II, B-I, C-III, D-IV

153. _____ is a psychological attachment to certain effects – such as euphoria.

- (1) Immunity (2) Addiction
(3) Syndrome (4) Allergy

154. Match List-I with List-II:

List-I

List-II

- (A) *cryIAc* (I) Mobile genetic element
(B) *cryIAb* (II) Transgenic crop
(C) Golden rice (III) Corn borer
(D) Transposons (IV) Cotton bollworms

Choose the **correct** answer from the options given below:

- (1) A-II, B-I, C-IV, D-III
(2) A-I, B-III, C-II, D-IV
(3) A-IV, B-III, C-II, D-I
(4) A-III, B-II, C-IV, D-I

155. Arrange the following steps involved in the activation of Bt toxin in insects:

- (a) Ingestion of inactive protoxin
(b) Binding of activated toxin to midgut epithelial cells
(c) Conversion of protoxin to active toxin in the insect's gut
(d) Creation of pores leading to cell lysis
(e) Death of the insect

Choose the correct sequence of events from the options given below:

- (1) (c)-(a)-(d)-(b)-(e)
(2) (a)-(c)-(b)-(d)-(e)
(3) (c)-(a)-(b)-(e)-(d)
(4) (a)-(b)-(c)-(d)-(e)

156. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: Cancer is a non-infectious disease.

Reason R: Cancer cannot be transmitted from one person to another.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) A is true but R is false.
(2) A is false but R is true.
(3) Both A and R are true and R is the correct explanation of A.
(4) Both A and R are true and R is NOT the correct explanation of A.

157. Given below are two statements:

Statement I: Gene therapy is an attempt to treat a person suffering from a hereditary disease.

Statement II: Adenosine deaminase is an enzyme crucial for the immune system to function.

In the light of the above statements, choose the *most appropriate* answer from the options given below:

- (1) Statement I is correct but Statement II is incorrect.
(2) Statement I is incorrect but Statement II is correct.
(3) Both Statement I and Statement II are correct.
(4) Both Statement I and Statement II are incorrect.

158. The foetus receives some antibodies from their mother, through the placenta during pregnancy. This is an example of:

- (1) active immunity.
(2) passive immunity.
(3) cell-mediated immunity.
(4) None of these

159. Given below are some statements (A-D).

- A. Two polypeptides of human insulin are linked together by hydrogen bonds.
B. The origin of replication is responsible for the copy numbers of the linked DNA.
C. The genes encoding resistance for antibiotics are selectable markers.
D. DNA is a hydrophobic molecule.

Choose the **correct** statements from the options given below:

- (1) A and D only
- (2) B and C only
- (3) A, B and C only
- (4) A, B, C and D

160. Which processes are utilized in biotechnological applications in medicine?

- (a) Gene therapy
- (b) Production of transgenic animals
- (c) Development of bio-pesticides
- (d) Recombinant DNA technology
- (e) Use of biological products in treatments

Choose the correct answer from the options given below:

- (1) (a), (d) and (e)
- (2) (b), (c) and (e)
- (3) (a), (b) and (d)
- (4) (c), (d) and (e)

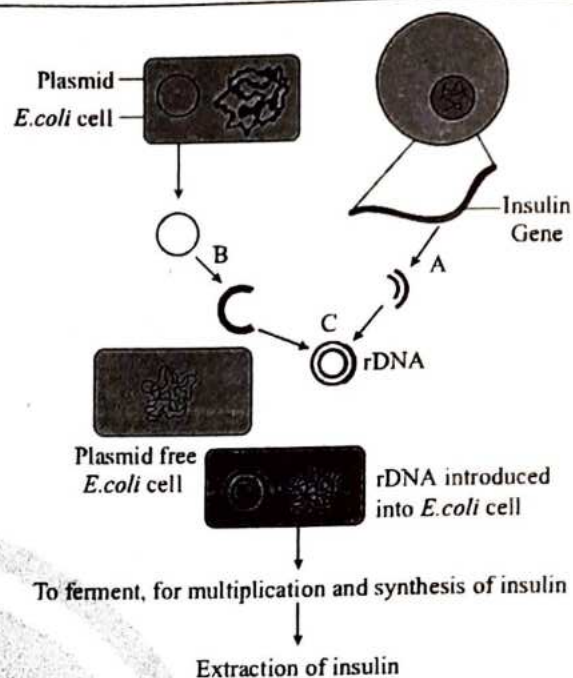
161. Match List-I with List-II:

List-I	List-II
(A) Bacterial disease	(I) Ringworm infection
(B) Fungal disease	(II) Malaria
(C) Viral disease	(III) Typhoid
(D) Protozoan disease	(IV) Common cold

Choose the **correct** answer from the options given below:

- (1) A-IV, B-III, C-II, D-I
- (2) A-I, B-III, C-II, D-IV
- (3) A-II, B-III, C-IV, D-I
- (4) A-III, B-I, C-IV, D-II

162. Following figures show the different steps involved in production of genetically engineered insulin. Identify the enzymes involved in steps A, B and C and select the correct statement(s) regarding this.



- (1) Enzymes involved in steps A and B are same, i.e., restriction exonucleases
- (2) Enzymes involved in steps B and C are same, i.e., restriction endonucleases
- (3) Enzymes involved in step C is DNA ligases
- (4) Both (1) and (3)

163. These questions consist of two statements each, printed as Assertion (A) and Reason (R). While answering these questions, you are required to choose any one of the following four responses.

Assertion (A): While performing autoradiography, the clone of cells having mutated gene will not appear on the photographic film.

Reason (R): The probe will not have complementarity with the mutated gene.

- (1) Both Assertion and Reason are True and the Reason is a correct explanation of the Assertion.
- (2) Both Assertion and Reason are True but Reason is not a correct explanation of the Assertion.
- (3) Assertion is True but the Reason is False.
- (4) Assertion is False but the Reason is True.

164. Which of the following is characteristic of the primary immune response?

- (1) Rapid activation of memory cells.
- (2) Slow production of antibodies.
- (3) It is of high intensity.
- (4) No memory cell formation.

165. Given below are two statements:

Statement I: Genetic engineering can overcome the drawbacks of traditional hybridisation.

Statement II: Genetic engineering can create desired DNA sequences to meet specific requirements.

In the light of the above statements, choose the *most appropriate* answer from the options given below:

- (1) Statement I is correct but Statement II is incorrect.
- (2) Statement I is incorrect but Statement II is correct.
- (3) Both Statement I and Statement II are correct.
- (4) Both Statement I and Statement II are incorrect.

166. Match the columns.

Column-I	Column-II
(a) Bt toxin	(i) Silencing of m-RNA
(b) RNA interference	(ii) ADA cDNA
(c) RDT	(iii) <i>cry</i> gene
(d) Gene Therapy	(iv) DNA or RNA probe

- (1) (a)-(i), (b)-(iii), (c)-(ii), (d)-(iv)
- (2) (a)-(iii) (b)-(i) (c)-(ii) (d)-(iv)
- (3) (a)-(iii) (b)-(i) (c)-(iv) (d)-(ii)
- (4) (a)-(ii) (b)-(i) (c)-(iii) (d)-(iv)

167. Given below are two statements:

Statement I: Reckless behaviour, vandalism and violence are immediate adverse effects of drugs and alcohol abuse.

Statement II: Excessive doses of drugs may lead to coma and death.

In the light of the above statements, choose the *most appropriate* answer from the options given below:

- (1) Statement I is correct but Statement II is incorrect.
- (2) Statement I is incorrect but Statement II is correct.
- (3) Both Statement I and Statement II are correct.
- (4) Both Statement I and Statement II are incorrect.

168. Which of the following statements is **correct** with respect to traditional hybridisation?

- (1) Procedure extensively used in plant breeding.
- (2) Often leads to inclusion of undesirable genes.
- (3) Involves selective inclusion and multiplication of desired genes only.
- (4) Enabled qualitative and quantitative improvement in food production.

169. Given below are some statements (A-D).

- A. The recombinant DNA technology has enabled mass production of safe and more effective therapeutic drugs.
- B. About 30 recombinant therapeutics have been approved for human use in the India.
- C. Management of adult-onset diabetes is possible by taking insulin at regular time intervals.
- D. Using conventional methods of diagnosis, early detection of diseases is possible.

Choose the **incorrect** statements from the options given below:

- (1) A and C only
- (2) A and B only
- (3) C and D only
- (4) B and D only

170. Significance of heat shock method in bacterial transformation is:

- (1) DNA is directly injected into the nucleus of an animal cell.
- (2) uptake of DNA using high velocity micro-particles of gold.
- (3) uptake of DNA through pores in bacterial cell wall.
- (4) expression of antibiotic resistance gene.

171. Match List-I with List-II:

List-I	List-II
(A) Physical barrier	(I) Macrophages
(B) Physiological barrier	(II) Skin
(C) Cytokine barrier	(III) Saliva from mouth
(D) Cellular barrier	(IV) Interferons

Choose the correct answer from the options given below:

- (1) A-III, B-II, C-IV, D-I
- (2) A-II, B-III, C-IV, D-I
- (3) A-III, B-I, C-IV, D-II
- (4) A-II, B-IV, C-I, D-III

172. Which of the following is correct for *Meloidegryne incognitia*?

- (1) A bacterium that can provide insulin.
- (2) A yeast used in production of vaccine.
- (3) A nematode that infects tobacco plant.
- (4) None of these

173. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion A: *E.coli* having pBR322 with DNA insert at *Bam*H I site cannot grow in a medium containing tetracycline.

Reason R: Recognition site for *Bam*H I is present in tetracycline region of pBR322.

In the light of the above statements, choose the correct answer from the options given below:

- (1) A is true but R is false.
- (2) A is false but R is true.

- (3) Both A and R are true and R is the correct explanation of A.
- (4) Both A and R are true and R is NOT the correct explanation of A.

174. Which of the following options can lead to neoplastic transformation?

- (1) Ionising radiations
- (2) Non- ionising radiations
- (3) Oncogenic viruses
- (4) All of these

175. Match List-I with List-II:

List-I	List-II
(A) Green Revolution	(I) Polio vaccines
(B) α -1- antitrypsin	(II) Rosie
(C) Transgenic cow	(III) Tripling the food supply
(D) Transgenic mice	(IV) Emphysema

Choose the correct answer from the options given below:

- (1) A-III, B-IV, C-I, D-II
- (2) A-IV, B-III, C-II, D-I
- (3) A-III, B-IV, C-II, D-I
- (4) A-II, B-IV, C-III, D-I

176. Given below are two statements:

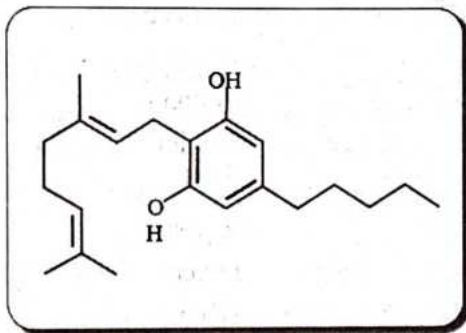
Statement I: In gel electrophoresis, agarose gel is used to separate DNA fragments.

Statement II: Selection of recombinants due to inactivation of antibiotics is a cumbersome procedure.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (1) Statement I is correct but Statement II is incorrect.
- (2) Statement I is incorrect but Statement II is correct.
- (3) Both Statement I and Statement II are correct.
- (4) Both Statement I and Statement II are incorrect.

177. Nicotine mainly stimulates which of the following glands?
 (1) Pineal gland (2) Thyroid gland
 (3) Thymus gland (4) Adrenal gland
178. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:
Assertion A: Gene therapy using lymphocytes from the blood of patient, is a permanent cure.
Reason R: Lymphocytes are not immortal, the patient requires periodic infusion of genetically engineered lymphocytes.
 In the light of the above statements, choose the correct answer from the options given below:
 (1) A is true but R is false.
 (2) A is false but R is true.
 (3) Both A and R are true and R is the correct explanation of A.
 (4) Both A and R are true and R is NOT the correct explanation of A.
179. While isolating DNA from bacteria, which of the following enzymes is **not** used?
 (1) Lysozymes
 (2) Ribonuclease
 (3) Deoxyribonuclease
 (4) Protease
180. Identify the given structure and choose the correct option.



- (1) Skeletal structure of cannabinoid molecule
 (2) Chemical structure of morphine
 (3) Skeletal structure of benzodiazepine molecule
 (4) Chemical structure of amphetamine molecule

181. The plasmid is:
 (1) linear RNA.
 (2) circular RNA.
 (3) linear DNA.
 (4) circular DNA.
182. Given below are some statements (A-D).
 A. The restriction enzyme acts as a molecular scissors.
 B. Presence of insert in β galactosidase gene, gives blue coloured colonies.
 C. The EFB stands for European Federation of Biology.
 D. In chemical engineering processes, it is important to maintain sterile conditions.
 Choose the **incorrect** statement(s) from the options given below:
 (1) A and D only
 (2) A and B only
 (3) B and C only
 (4) D only
183. α interferons are primarily used in treatment of:
 (1) AIDS.
 (2) malaria.
 (3) cancer.
 (4) typhoid.

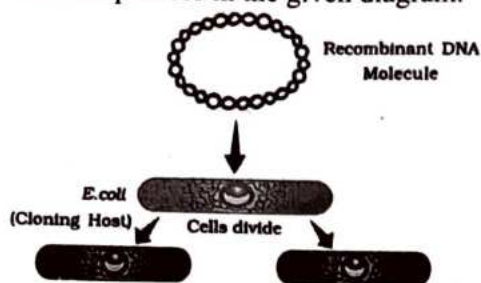
184. Match List-I with List-II:

List-I	List-II
(A) Lepidopterans	(I) ssRNA tagged with radioactive molecule
(B) Dipterans	(II) Beetles
(C) Coleopterans	(III) Flies
(D) Probe	(IV) Armyworms

Choose the **correct** answer from the options given below:

- (1) A-IV, B-III, C-II, D-I
 (2) A-IV, B-III, C-I, D-II
 (3) A-III, B-IV, C-II, D-I
 (4) A-II, B-IV, C-III, D-I

185. Name the process in the given diagram.



- (1) Spooling
- (2) Elution
- (3) Transformation
- (4) Polymerase chain reaction (PCR)

Zoology : Section-B (Q. No. 186 to 200)

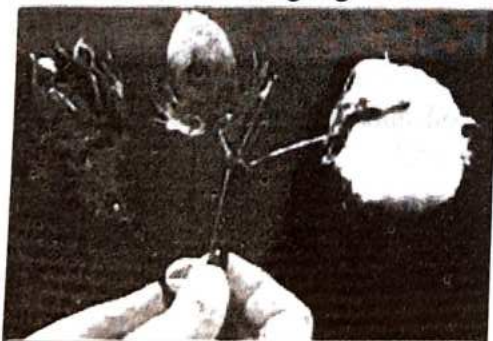
186. Symptoms of allergy can be reduced by:

- (1) adrenaline. (2) steroids.
- (3) retroviral drugs. (4) Both (1) and (2)

187. Which of the following statements is **incorrect** regarding genetically engineered insulin?

- (1) *E. coli* is used in producing genetically engineered insulin.
- (2) Chain A and chain B of insulin are formed separately.
- (3) First genetically engineered insulin was prepared by an American company Eli Lilly.
- (4) Genetically engineered insulin have C peptides.

188. In the given diagram, the plant is infected by which of the following organisms?



- (1) Corn borer
- (2) Cotton bollworm
- (3) *Bacillus thuringiensis*
- (4) *Meloidogyne incognita*

189. What is the role of memory cells in acquired immunity?

- (1) To produce histamine during allergic reactions.
- (2) To remember and respond more rapidly to pathogens upon subsequent exposures.
- (3) To destroy pathogens through phagocytosis.
- (4) To form the physical barriers against pathogens.

190. How many statements are correct regarding the vaccine safety?

- (a) Transgenic mice are being developed for use in testing the safety of vaccines.
- (b) Vaccines cannot be given directly to humans as it can be fatal so they should be tested on other organisms for safety check.
- (c) Before a vaccine is released into the market and qualified for use in human trials, it is extensively tested for safety and effectiveness in transgenic cats.
- (d) The transgenic mouse model organisms are least applicable.

Select the correct option:

- (1) One (2) Two
- (3) Three (4) Four

191. Given below are two statements:

Statement I: The *in vitro* fertilisation leading to 'test-tube' baby is a technique included under biotechnology.

Statement II: Making multiple identical copies of any template DNA, is called cloning.

In the light of the above statements, choose the *most appropriate* answer from the options given below:

- (1) Statement I is correct but Statement II is incorrect.
- (2) Statement I is incorrect but Statement II is correct.
- (3) Both Statement I and Statement II are correct.
- (4) Both Statement I and Statement II are incorrect.

192. Match List-I with List-II:

List-I	List-II
(A) Sedative	(I) Cocaine
(B) Sense of euphoria	(II) Cannabinoids
(C) Treats insomnia	(III) Morphine
(D) Abused by sportsmen	(IV) Barbiturates

Choose the **correct** answer from the options given below:

- (1) A-IV, B-III, C-II, D-I
- (2) A-III, B-II, C-I, D-IV
- (3) A-II, B-III, C-IV, D-I
- (4) A-III, B-I, C-IV, D-II

193. Select the **correct** statement.

- (1) The polymerase enzyme joins the gene of interest to the vector DNA.
- (2) Restriction endonuclease works on the principle of removing the nucleotides, at specific sequences from the DNA molecule.
- (3) Spooling is used for amplification of gene of interest.
- (4) Small scale production of recombinant protein can be done from bioreactors.

194. These questions consist of two statements each, printed as Statement-I and Statement-II. While answering these questions, you are required to choose any one of the following four responses.

Statement-I: Totipotency is an artificially induced phenomenon.

Statement-II: The mammalian gametes is also known to possess totipotent traits.

- (1) Both Statement-I and Statement-II are correct.
- (2) Both Statement-I and Statement-II are incorrect.
- (3) Statement-I is correct & Statement-II is incorrect.
- (4) Statement-I is incorrect & Statement-II is correct.

195. Match List-I with List-II with respect to the nomenclature of restriction enzyme:

List-I	List-II
(A) <i>E</i>	(I) Order of isolation from strain of bacteria
(B) <i>co</i>	(II) Name of strain
(C) <i>R</i>	(III) Name of genus
(D) <i>I</i>	(IV) Name of species

Choose the **correct** answer from the options given below:

- (1) A-IV, B-III, C-II, D-I
- (2) A-III, B-IV, C-II, D-I
- (3) A-II, B-III, C-IV, D-I
- (4) A-III, B-I, C-IV, D-II

196. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: Vaccination is an effective way to prevent diseases.

Reason R: Vaccine stimulates the immune system to produce antibodies against specific pathogens.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) A is true but R is false.
- (2) A is false but R is true.
- (3) Both A and R are true and R is the correct explanation of A.
- (4) Both A and R are true and R is NOT the correct explanation of A.

197. Sporozoites of the malarial parasites are found in:

- (1) the saliva of male *Anopheles* mosquito.
- (2) the saliva of infected female *Anopheles* mosquito.
- (3) RBCs of an infected human.
- (4) the spleen of an infected human.

198. Given below are some statements (A-D).

- A. The first restriction enzyme to be isolated and characterised was *Hind* II.
- B. Stanley Cohen and Herbert Boyer in 1972, constructed the first recombinant DNA.
- C. Restriction enzymes belong to larger class of enzymes called nucleases.
- D. Primers are small chemically synthesised oligonucleotides that are complementary to the regions of DNA.

Choose the **correct** statements from the options given below:

- (1) A and D only
- (2) A and B only
- (3) B, C and D only
- (4) A, B, C and D

199. Given below are two statements:

Statement I: Polymerase chain reaction (PCR) is now routinely used to detect HIV in suspected AIDS patients.

Statement II: The time-lag between the infection and the appearance of AIDS symptoms may vary from a few months to 5–10 years.

In the light of the above statements, choose the *most appropriate* answer from the options given below:

- (1) Statement I is correct but Statement II is incorrect.
- (2) Statement I is incorrect but Statement II is correct.
- (3) Both Statement I and Statement II are correct.
- (4) Both Statement I and Statement II are incorrect.

200. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R:

Assertion A: The bone marrow is a secondary lymphoid organ.

Reason R: The secondary lymphoid organs provide the sites for interaction of lymphocytes with the antigen.

In the light of the above statements, choose the **correct** answer from the options given below:

- (1) A is true but R is false.
- (2) A is false but R is true.
- (3) Both A and R are true and R is the correct explanation of A.
- (4) Both A and R are true and R is NOT the correct explanation of A.